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Authoritarian but responsive: Local regulation of industrial energy efficiency in Jiangsu, China

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Abstract

Regulatory behavior and effectiveness in authoritarian settings are subject to alternative characterizations. By tracing enforcement processes through a variety of case studies, this article proposes and refines a new model, at least with respect to energy efficiency regulations in China: *authoritarian but responsive*. Local rulemaking and operationalization is authoritarian, with strong and coordinative bodies of regulation, strategic plans, and active involvement of local authorities. Local authorities, however, often find themselves facing a welter of laws imposed on companies that create competing priorities for these local officials who then must struggle to find pragmatic solutions. On numerous occasions, such satisficing behavior by local officials makes them responsive to the performance and demands of regulated firms. Embedded in the decentralized authoritarian context, the *authoritarian but responsive* approach is found to be a rational choice of local governments and different from previous conceptualizations. It helps local governments coordinate across a diverse array of regulatory issues. Drawing on environmental enforcement in China, the model provides consistent explanations for the seemingly changing and discretionary enforcement incidents.

Keywords: Chinese energy policy, decentralized authoritarianism, policy implementation, regulatory enforcement.

1. Introduction

The need for effective implementation and enforcement of environmental policies is a common issue globally and of particular salience in rapidly industrializing economies. Environmental policies are essentially challenged by the fact that a firm may rationally choose to evade rather than comply if the evasion benefits are found to outweigh the potential costs of regulatory punishment (Gray & Shimshack 2011). Governments, too, may fail to enforce their own policies because of their institutional shortcomings (Schofer & Hironaka 2005). With increasing demand to reduce pollution and energy consumption, recent research in China indicates a regional variation in regulatory effectiveness. There is general improvement in implementation and enforcement across the country over time (van Rooij & Lo 2010; van Rooij *et al.* 2013), and coastal areas practice stronger enforcement than the inland west (Schwartz 2003; van Rooij & Lo 2010).

Changing regulatory effectiveness is attributed to three different sources in the literature. The first concerns institutional settings, such as the coming of "authoritarian environmentalism" (Beeson 2010), central government support (Tilt 2007; Zhan *et al.* 2014), and the convergence of multiple institutional forces (van Rooij & Lo 2010). The second focuses on the status of actors involved in the regulatory process, including the autonomy and strength of regulatory agencies (Schwartz 2003; Lo & Tang 2006; Liu *et al.* 2015), roles and perspectives of regulators (Chan *et al.* 1995; Lo *et al.* 2006; Eaton & Kostka 2014), bargaining power of regulated industries (Wang *et al.* 2003), and civil society support (Schwartz 2003; Tilt 2007; Tang *et al.* 2010; van Rooij *et al.* 2013). Finally, the third source draws on the specific strategies and styles that regulatory authorities use in enforcement, such as

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campaign-style enforcement (Liu *et al.* 2015), which strengthens the capacity and priority for enforcing environmental policies, and also the reliance on more formal approaches at the street level (Lo *et al.* 2009).

While the explanations originate from seemingly different perspectives, scholars have started to link the first example, institutional setting, with the third source, the behaviors of regulators. China's institutional system is portrayed in the literature as having a dual nature. The often-stressed authoritarian system of top-down political control, limited participation, and paternalism is increasingly recognized and is the first side of the duality (Landry 2008; Xu 2011). The other half that complements authoritarianism includes not only economic liberalization and resilience (Roy 1994; Nathan 2003), but also decentralization (Landry 2008, Xu 2011), fragmentation (Mertha 2009), consultation (Teets 2013), and even political deliberation (He & Warren 2011).

The institutional setting shapes political and promotional incentives of local governments (Li & Zhou 2005), which leads both to their competition and policy experimentation (Heilmann 2008), and to their collaboration and collusion (Zhou 2010). Particularly for environmental and climate regulations, some authors argue that local governments take authoritarian, non-participatory approaches, which produce immediate policy outputs but not necessarily intended outcomes (Gilley 2012; Kostka 2015). This is accompanied by scholarly observations of pluralized environmental actors (van Rooij *et al.* 2014) and their extra-legal behaviors intentionally ignored by the regulators (Weller 2012), as well as a call for broader, decentered regulation (Johnson 2016).

This study takes a further step toward understanding the behavior and impact of local government in policy implementation, embedded in a regionally decentralized authoritarian regime. In contrast with considering enforcement as a one-time decision, we focus on the longer-term interactive process between regulators and regulated industrial firms in China, where demands for environmental protection and climate mitigation are an increasingly important dimension of local governments' administrative concerns. By tracing the process of regulatory interactions, we try to develop a more consistent, context-independent conceptualization of regulatory behaviors and impacts.

We trace the local implementation and enforcement process in the case of China's energy efficiency policy based on interviews and archival documents from Jiangsu Province. The focus on a single but comprehensive regulatory area helps us build and test our conceptualization model and explanations. Since the end of the 10th Five-Year Plan (FYP) period of 2005, the national and provincial governments of China have been seriously dedicated to reducing energy consumption per unit of gross domestic product (GDP) with punitive and exacting measures. The policies have featured compulsory targets for national energy efficiency in the 11th and 12th FYPs, disaggregation and delegation of the targets through administrative ladders with spatial differences, and use of multiple policy instruments and energy saving measures (Zhou *et al.* 2010; Price *et al.* 2011; Zhu 2013). Jiangsu Province has one of the most developed economies and most stringent energy efficiency targets in China.

Our findings indicate that local governments, although dominant in rulemaking and operationalization, also respond to the proposals, demands, and performance of regulated firms by preferentially lowering enforcement levels or alleviating previous punitive enforcement in cases when firms, attempting to respond to a welter of sometimes competing regulatory dicta, become willing to offer solutions that contribute back to local communities and governance. The *authoritarian but responsive* approach is found both to lead to better understanding and cooperation by regulated firms and to improve government coordination across regulatory issues. Embedded in the institutional setting, the authoritarian but responsive approach is found here to be a rational and consistent choice by local government.

As descriptive and explanatory rather than prescriptive, the authoritarian but responsive model we identified contributes to an understanding of regulatory behaviors and associated impacts within decentralized authoritarian settings. It connects the ideal of responsive regulation with institutional logic and reality in China, showing why and how the government responds and what differences it makes. Seemingly changing or differential enforcement follows a consistent rationale for better governance through responsiveness and coordination. This approach has advantages in local governance over the inflexible, top-down environmental authoritarianism that often creates ineffective policy outcomes (Gilley 2012; Kostka 2015) and highly variable enforcement styles selected by street-level bureaucrats whose actions confuse and subsequently undermine understanding and compliance by the regulated community (May & Wood 2003). It also differs from the theories of responsive authoritarianism (Weller 2012) and responsive regulation (Ayres & Braithwaite 1992; Parker 2013) in its distinctive scale and scope of regulatory responsiveness.

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2. Institutional background of energy efficiency policy

Institutional settings shape the behaviors of regulatory agencies and particularly their responsiveness (Nielsen 2006). On the other hand, regulatory approaches may complement and strengthen the limited capacity of a regulatory state (Braithwaite 2006). While normative arguments for certain regulatory approaches are well developed, there is also a necessary role for empirical investigation, especially for institutions undergoing rapid changes. Herein, we first review the literature of institution and regulation, specifically regarding variation in enforcement, then introduce energy efficiency policies in China, and finally, propose our conceptualization for further development based on the literature and new empirical findings.

2.1. Institutional context and environmental regulation

Scholars have described regulation in China as a regionally decentralized authoritarian system (Landry 2008; Xu 2011): it features a combination of centralized political and personnel controls, and regionally-based economic and administrative governance, where local governments compete with one another in economic development and policy reforms. The dual nature of the institution further manifests itself with respect to a few aspects of authoritarianism, including being formal rational, administratively competitive, fragmented, guided experimental, and responsive (van Rooij 2014).

Environmental regulation operates within the decentralized authoritarian system. A theoretical approach that explains how governments in an authoritarian context respond to environmental challenges is authoritarian environmentalism. It is characterized by the primacy of capable, uncorrupted executive authority on improving environmental outcomes, technocratic decisionmaking, limited public participation, and suppressed individual liberty with regard to environmentally destructive behaviors (Beeson 2010; Gilley 2012). This model is considered to have both benefits and shortcomings. The benefits of authoritarian environmentalism include progressive policy response even under complex political pressures (Beeson 2010), resource mobilization, and power redistribution that facilitates policy implementation (Liu *et al.* 2015). On the negative side, rapid responses in policymaking may not necessarily lead to effective implementation around desired environmental outcomes (Gilley 2012), especially when frequent leadership turnovers encourage local governments to adopt shorter-term approaches (Eaton & Kostka 2014).

Regional decentralization also serves as a basis for pluralized policy processes and varying enforcement. To be responsive to diverse demands, barriers to entry in agenda setting and policymaking are lowered (Wang 2008; Mertha 2009). In addition to direct regulators, actors engaged in the policy process may include business associations (Yu *et al.* 2014), local media (Chin 2011), think tanks (Zhu 2009), and government officials peripheral to the focused policies (Mertha 2009). In particular, environmental regulation features a broader incorporation of the court (Zhang 2016), prosecutors (Shi & van Rooij 2016), urban residents, and journalists (Johnson 2016).

While the pluralized policy actors may explain part of the enforcement variation, the major factor is differences among local regulatory authorities that are the dominant actors in regional economic and administrative governance. Differences in these authorities occur across jurisdictions in fiscal strength (Schwartz 2003), human resources (Schwartz 2003; Wheeler 2005), perception of enforced policies (Lo *et al.* 2006), environmental values (Chan *et al.* 1995), and the policies implemented in accord with local interests and policy priorities (Qi *et al.* 2008; Kostka & Hobbs 2012). These differences sometimes converge with other social and economic forces, and lead to variation in enforcement by regulatory authorities (van Rooij & Lo 2010).

Previously, enforcement of centrally designed environmental policies was typically considered to be more or less half-hearted and ineffective because of the conflict between the economic costs of regulation and the local priority of economic growth. Recent energy efficiency policies, however, have been effectively implemented within the provinces without damaging the economy. The process of energy saving was achieved mainly through technological change rather than industrial relocation: unlike the conventional wisdom that regulated industries would move out of areas of higher regulation, they invested substantially in more efficient technologies instead (Zhu & Ruth 2015). The policies are reviewed further, followed by preliminary explanations.

2.2. Energy efficiency policies

Energy conservation became part of the policy agenda in China in the 1980s. Since then, energy intensity, measured as energy consumption per unit GDP, decreased for about 20 years, mainly driven by technological

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upgrading (Zhang 2003; Ma & Stern 2008). This decline, however, was suddenly reversed during 2002–2004, creating the need for new policies. Facing these challenges, the central government established and refined a comprehensive and ambitious policy framework for the more efficient use of energy, especially in the industrial sector. For research purposes, the post 2004 period became a distinctive boundary from which to measure policy-driven change.

The policy framework consisted of general guidelines that proposed energy saving measures (Table A1), and detailed policies and programs to implement those measures (Table A2). General guidelines include:

1. *Energy Conservation Law Amendments* and the *11th and 12th Five-Year Plans for National Economic and Social Development* ratified by the National People's Congress.
2. Energy saving or climate mitigation decisions and plans announced by the State Council.
3. Specified plans for energy saving and climate mitigation in the industrial sector and longer-term plans designed by ministries in the State Council.

The most important change called for the quantification and application of compulsory targets for reducing national energy intensity by 20 percent between 2006 and 2010, and by another 16 percent between 2011 and 2015. The key regulatory agencies in policy design and implementation are the national and local Development and Reform Commissions, which are responsible for guiding economic and social development, and the Ministry of Industry and Information Technology and its local delegations, which are in charge of policies for individual industries.

Collectively, these guidelines propose multiple industrial energy saving measures, including: target-setting for energy consumption per unit GDP, target disaggregation, and examination at subnational administrative levels; energy-efficiency performance standards for industrial processes and products; administration for firm-level energy-saving assessments, target-setting, planning, and supervision; promotion for energy-saving innovation, information, and technology adoption; industrial structural adjustment and elimination of outdated production capacity; and policy reforms to support energy saving (Zhu & Ruth 2015). Because industries feature different levels of energy intensity and excessive production capacity, more policy emphasis is attached to those industries with higher energy intensity and overcapacity in the guidelines.

Detailed policies and programs were designed to focus on single or multiple energy saving measures, and single or multiple policy instruments. These instruments can be mandatory, voluntary, incentive-based, or derived from information disclosure. A major firm-level program, for example, is the Top-1000 Enterprise Energy Saving Program: firms with annual energy consumption beyond a threshold were required to join the program for energy saving, but their targets and planning for energy saving were proposed voluntarily and then negotiated with local governments. The program also incorporated information disclosure and incentives. The elimination of outdated production capacity became mandatory, providing information on which to base stricter enforcement specifying that facilities below certain technical standards must be retired following a timeline; the outdated facilities were charged with higher electricity prices – an incentive-based instrument – before retirement.

2.3. An authoritarian but responsive approach for energy efficiency?

Energy efficiency policies have provided an opportunity for exploring the behavior of local government in policy implementation and enforcement within the regionally decentralized authoritarian regime (Kostka & Hobbs 2012; Eaton & Kostka 2014; Liu *et al.* 2015). Compared to traditional environmental issues, industrial energy efficiency is of relatively less interest to nonconventional policy actors in China, such as the media and the public. Therefore, we can focus on the local regulatory authorities when studying policy enforcement.

One can expect an authoritarian approach to local regulation for energy efficiency. The national policy design was determined in a short period with little participation from outside the government. Overall policy implementation was manifested by the disaggregation and delegation of energy efficiency targets in a top-down manner through the administrative ladder. Fulfillment of the energy efficiency target was linked to top leaders' evaluation and promotion in a region (Zhou *et al.* 2010), creating strong political motivation to ensure policy compliance.

There is, however, also reason to anticipate a responsive part of the local regulation. The participation of non-regulatory actors has been observed as an important force in policymaking (Mertha 2009) and enforcement

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in China (Schwartz 2003; van Rooij *et al.* 2013). Governments also intentionally leave more room for the media (Stockmann 2013) and civil society (Weller 2012), to respond to public demand (Chen *et al.* 2016), and to collaborate with civil society to promote the delivery of public welfare (Teets 2013). Specifically, government policy choice is affected by firms' strategic actions (Child & Tsai 2005; Zhu & Chertow 2017) and dominance in the local economy (Lorentzen *et al.* 2014), some of which is carried out by large energy consumers.

We are interested not only in the existence and appearance of regulatory responsiveness, but also in whether it contributes to a better regulatory model in China, both prescriptively and descriptively. The authoritarian rule is complemented with deliberative practices, which is considered to affect political development (He & Warren 2011). As argued by Ayres and Braithwaite (1992), prescriptively, responsive regulation is a democratic ideal and has most often led to a dynamic process of enforcement and improved compliance by firms. In a descriptive and explanatory manner, we also expect to find a more consistent rationale behind the seemingly random variation in individual enforcement incidents.

3. Methods

To develop and refine explanations of local government environmental policy enforcement behavior, we apply a case study framework for analytic generalization. The case studied is government enforcement of energy efficiency policies in Jiangsu Province; the unit of analysis is an enforcement process at a single firm, which may involve regulatory authorities at different levels in Jiangsu and their interactions with the firm through time. Data were collected through interviews and archival documents of enforcement incidents in 21 firms from 2012 to 2015. Firm names and locations are coded to preserve anonymity. Analytic generalization is completed based on strategies of descriptive explanations and comparison across units of analysis (Yin 2014). Through this process, the preliminary explanation is revised and refined to provide analytic explanations for government behaviors in policy enforcement.

3.1. Case selection

Jiangsu Province was selected because it has the largest industrial sector among Chinese provinces, with well developed economic prosperity; its energy consumption per GDP was 29 percent lower than the national average in 2010 (People's Government of Jiangsu 2012), but it was still one of the provinces with the largest percentage target for energy intensity reduction (State Council 2011). By considering a single province, we avoid the potential variation among provinces that may complicate policy enforcement, for example in their individual cultures, interests, and priorities; their negotiation with the central government; and interpretation of the policies. Much variation may still exist for identifying and generalizing enforcement behaviors among municipalities and lower-level jurisdictions, especially between the more developed south and less developed north within the province.

Energy efficiency became important in Jiangsu not only because of the policy itself, but also in the context of other policy targets. Two former Chinese Presidents, Jiang Zemin and Hu Jingtao, both required Jiangsu to be the first province to fully build a "well-off society" (*Xiaokang Shehui*) and to complete modernization (People's Government of Jiangsu 2014), two policy priorities of the central government with the former to be reached by 2020 and the latter by 2050 nationally. The goals of the provincial government, based on its self-designed sets of progressive qualification standards, were to reach a well-off society in each county of the province by 2015 and modernization for the province as a whole by 2020 (China News 2011). Energy consumption per unit GDP is included in both sets of qualifications, at 0.062 kg of coal equivalent per RMB for the former and 0.05 kg of coal equivalent per RMB for the latter (Jiangsu Development and Reform Commission 2013a,b). Because the government of Jiangsu Province has both strong motivation and a social and economic basis to seriously enforce the policy, this enables a focus on the interacting behavior in policy enforcement within the context of decentralized authoritarianism.

We focus on enforcement in four municipalities in Jiangsu Province, characterized in Table 1 with a qualitative comparison that prevents their identities from being revealed. Three are in the south of the province with highly industrialized or post-industrialized economies, a high level of socioeconomic development, and strong political interest in reducing overall energy intensity within their jurisdictions. Plummeting energy intensity not

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Table 1 Characteristics of the four municipalities

City	A	B	C	D
Energy intensity (consumption per GDP)	High	Medium	Medium	High
Actual energy intensity reduction 2005–2010	High	Medium	Medium	Low
Energy intensity reduction target 2011–2015	High	Medium	High	Medium
Actual energy intensity reduction 2011–2014	High	High	Medium	Medium
Socioeconomic development level	High	Medium	High	Low
Industrial share of GDP	Low	High	High	Medium

The characterization is relative to the average level in Jiangsu Province. GDP, gross domestic product.

only helps local administrative leaders with their career evaluation directly, but also indirectly as an important qualifying condition to move regions under their jurisdictions ahead of the provincial agenda for the "two policy priorities." To improve comparison and generalization, one municipality in the north was also selected, with a relatively lower level of socioeconomic development and a lower priority for energy efficiency. The municipalities may vary in their own policy agendas and administrative approaches.

Analytic generalization requires enforcement incidents to be selected based on theoretical replication, rather than representative sampling (Yin 2014). The incidents for analyses, therefore, were carefully selected to highlight contrasting contextual and internal factors that may affect interaction between the regulators and the regulated. Characteristics to be considered for the regulated include industrial sector, firm size, ownership, and age.

We focus on policy implementation and enforcement in four industries: cement, iron and steel, paper, and chemical products (Table 2). The industries feature different national-level and provincial-level regulations. Cement and steel industries, both experiencing serious overcapacity, have been strictly mandated to eliminate production facilities, especially those using certain technologies and under certain capacities, and then must stay below a capacity cap in each region. There are some requirements for specific facility elimination in the paper industry, but no explicit constraint for production capacity. The chemical industry has been required to eliminate outdated facilities, but only for a few processes that experience overcapacity, such as the chloral-alkali process. Firm size, ownership, and age may also affect the enforcement process. Larger firms are expected to have greater capability for energy saving, be more visible in enforcement, but also to have greater bargaining power.

Table 2 Selected industries and their characteristics

Industry	Cement[†]	Steel[‡]	Paper[§]	Chemical[¶]
National				
Production in 2005 (million ton)	1,070	350	32.0	—
Production in 2010 (million ton)	1,880	630	92.7	—
Industry-wide capacity closure achieved 2006–2010 (million ton)	340	195[††]	10	—
Capacity closure target by industry 2011–2015 (million ton)	250	96[††]	10	—
Reduction of energy consumption per product weight 2006–2010	12%[‡‡]	12.8%[§§]	18%	—
Reduction target of energy consumption per product weight 2010–2015	—	4%[§§]	22%	—
Energy intensity reduction 2006–2010	—	—	—	35.8%[¶¶]
Energy intensity reduction target 2011–2015	—	18%	—	20%[¶¶]
CO_2 emission intensity reduction target 2011–2015	17%	18%	—	17%[¶¶]
Jiangsu				
Industry-wide capacity closure achieved 2006–2010 (million ton)[a]	33.5	11.6[††]	0.5	—
Capacity closure target by industry 2011–2015 (million ton)[b]	15.0	0.02[††]	0.3	—

[†]Cement industry development plan during the 12th Five-year Plan (FYP). [‡]Iron and steel industry development plan during the 12th FYP. [§]Pulp and paper industry development plan during the 12th FYP. [¶]Petrochemical and chemical industry development plan during the 12th FYP. [††]Combined capacity for iron production and steel production. [‡‡]Only for cement clinker. [§§]Only key firms for statistical purpose are included. [¶¶]Combined reduction in the petrochemical and chemical industries. [a]Jiangsu energy-saving plan during the 12th FYP. [b]Jiangsu's planned closure of industrial outdated capacity during the 12th FYP.

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Ownership is expected to be associated with political connections and affect whether and how firms are treated preferentially in local regulation, as the literature suggests (Christmann & Taylor 2001; Wang & Jin 2007; Wu *et al.* 2012). Firms built in different periods are also considered to reflect the impact of capital vintage on energy efficiency and policy enforcement.

3.2. Data collection and analysis

Data of both the regulators and the regulated firms were collected mainly from two sources – archival documents and in-depth interviews – from 2012 to 2015. For the regulators, the main focus is local government subsidiaries in charge of industrial energy efficiency – the economic and information technology commissions, as well as development and reform commissions and environmental protection bureaus that also play different roles in managing industrial energy efficiency. Government websites and media provided general descriptions of the interpretation and implementation of energy efficiency and related policies by provincial and local authorities. The information was first collected by searching the websites and through search engines according to the key words of the policy programs listed in the Appendix. A thorough review of the online archives of the regulatory authorities followed, seeking documents directly related to energy efficiency regulation at the provincial and local levels. Open-ended interviews were conducted with two provincial and nine local officials in charge of energy saving and related areas to elicit their comments on major policies, policy interpretation, implementation, and effectiveness. The interviews were conducted in their own private offices.

With respect to the entities being regulated, background information of the selected firms was first collected in a screening process through their websites, media coverage, and government archives. Thirty-two semi-structured interviews were conducted at 21 firms, with their chief executives, directors in charge of production, energy management, or environment and safety personnel. Twenty-three interviews were permitted to be audio-recorded, and detailed notes were taken for the rest. Interviews were all conducted in Mandarin Chinese and usually lasted for one hour in the interviewees' offices or meeting rooms. An interviewee was first asked to briefly introduce the firm, its position in the industry, and detail all of its past energy saving initiatives. The interviewee

Table 3 Characteristics of firms interviewed

Firm	Municipality	Ownership	Number of employees	Cooperating subsidiary	Age (years)	Interviewees	Records
Cement 1	A	Private	250	No	30	1	Audio
Cement 2	A	State	1,000	Yes	40	2	Audio
Cement 3	A	Foreign	300	Yes	5	2	Notes
Cement 4	B	Private	800	No	12	3	Audio
Cement 5	B	Foreign	800	Yes	20	2	Audio
Cement 6	D	State	800	Yes	15	1	Audio
Cement 7	D	Private	300	No	35	1	Audio
Steel 1	A	Private	5,500	No	55	3	Audio
Steel 2	B	Private	400	No	12	2	Notes
Steel 3	C	State	4,000	No	55	2	Audio
Steel 4	D	Private	2,000	No	5	2	Notes
Paper 1	B	State	1,400	No	65	1	Audio
Paper 2	B	Foreign	5,600	Yes	20	1	Notes
Paper 3	C	Foreign	300	Yes	18	2	Audio
Paper 4	C	Foreign	550	Yes	20	1	Notes
Paper 5	D	Private	500	No	35	1	Audio
Chemical 1	A	Foreign	20	Yes	10	1	Audio
Chemical 2	B	State	4,300	no	55	1	Audio
Chemical 3	B	State	1,700	no	65	1	Notes
Chemical 4	B	Private	700	no	35	1	Audio
Chemical 5	B	Private	400	no	10	1	Audio

A–C represents three municipalities in the south and D represents a municipality in the north of Jiangsu Province.

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was then encouraged to describe the process supporting each initiative in detail, to explain the reasons and determinants for the initiative, and, particularly, to elaborate on any interaction with regulatory authorities in the process. Finally, the interviewee was asked to comment on their general relationship with the government.

The information collected was transcribed, coded, and triangulated between different data sources. For each enforcement incident, we first coded the interaction between the regulated firm and regulators in a temporal manner according to the engaged regulatory authorities, their enforcement levels, firm responses and compliance, and final results. We then coded the internal and contextual factors that affected the decisions of the regulators and the regulated for the same enforcement. When a statement was not clear enough for coding, follow-up calls with the interviewees were made. After descriptive explanations were developed for each enforcement incident, we generalized explanations across incidents by identifying similar patterns, comparing between alternative explanations, and aggregating concepts into more abstract ones. Our main focus was enforcement incidents at cement and steel firms, which were embedded in more diverse contexts and were under stricter policies for energy efficiency (Table 3).

4. Findings

We identified authoritarian rulemaking and operationalization by local governments that oversee industrial energy efficiency, followed by responsiveness in individual instances of enforcement according to firms' performance and proposals, which led to stringent but flexible enforcement. The scope of responsiveness varied across municipalities and through time, reflecting governments' coordination among locally important, but sometimes overlapping areas of governance. The general mechanisms are illustrated in Figure 1, and are explained in more detail.

4.1. Authoritarian rulemaking and operationalization

Under the general policy framework set by the central and provincial governments, local governments dominated in building regulatory capacity, designing energy saving strategies, and employing multiple policy instruments. Local rulemaking and operationalization follows the institutional logic and the model of authoritarian environmentalism, with a clear overarching goal of energy saving; direct, hierarchical decisionmaking inside the government; and little involvement of the target group of regulated firms or the general public.

Regulatory capacity for energy efficiency was strengthened at the local level with strong political power and coordination. All four municipal governments formed leading groups on energy saving or both energy saving and emissions reduction. A mayor or deputy mayor of the municipalities led these groups and involved chief

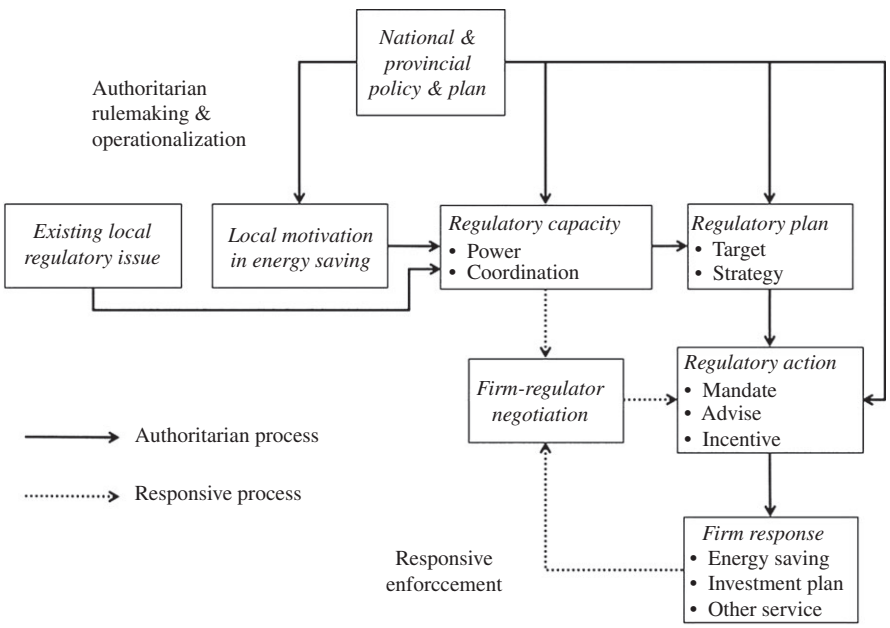

Figure 1 Authoritarian rule and responsive enforcement of local energy efficiency.

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officers from most of the municipal administrative bodies as members. The common members included (but were not limited to) chief officers from the Economic and Information Technology Commission (EITC), the Development and Reform Commission, the Environmental Protection Bureau, the Housing and Urban-rural Development Commission, the Science and Technology Bureau, the Transportation Bureau, the Land and Resources Bureau, and bureaus covering statistics, finance, state and local taxation, and education, as well as the propaganda department of the Party Commission.

Within each of the four municipalities, there are multiple counties and districts under municipal jurisdiction. These lower level governments were also required by the municipal governments to form similar leading groups. While there was an energy saving division in the EITC dedicated to regular management of industrial energy saving, the leading group was considered by the government interviewees to be of paramount significance: the leading group helped to prioritize energy saving in the local policy agenda, build political power for enforcement, coordinate between administrative bodies, and consolidate fiscal strength and human resources.[1] Because of the broad involvement of government officials from different agencies, favoritism toward specific firms according to personal or organizational connections, if any, was greatly mitigated.[2]

Moving toward implementation, each of the four municipal governments designed its own energy saving *plans and strategies*, not through negotiation with local industries but rather through a combination of following central and provincial plans and incorporating local knowledge. Some strategies varied across municipalities, reflecting regulators' knowledge about the status of local industries and potential directions for energy saving.[3] For example, industrial structural change was prioritized in some municipalities with higher shares of energy-intensive industries,[4] while improving the management of energy use in large firms was the main focus of municipalities with greater shares of energy consumption from a few major energy consumers.[5]

Some other strategies were commonly shared, following the priorities in central or provincial plans. One example is the setup of targets and responsibilities: municipal energy saving targets were announced, often more stringent than the ones negotiated with the provincial government to make sure the negotiated ones could be reached; the targets were required to be further disaggregated to lower level governments and to be part of the performance evaluation of the governments and their leaders. Another example is the emphasis on procedures for firm-level energy statistics and reporting, which led to the establishment of a center for energy conservation supervision in each of the four municipalities.[6]

Local authorities not only built strong regulatory capacity and identified strategies and measures, but also actively used multiple *policy instruments* and were deeply involved in the *implementation processes.* Table 4 presents detailed policy instruments where enforcement was found to be effective in facilitating decisionmaking or shaping the choices of energy saving in a given firm. While many of the policies applied generally to most of the firms, they were not listed if they were not considered important by the interviewees for their firms' energy saving. Examples include: (i) an energy saving agreement that was easily fulfilled with the targeted saving far below the actual saving; (ii) investment subsidies found to be trivial to the total investment and their availability not affecting firms' decision to invest; and (iii) energy efficiency standards that were below firms' performance, even without energy saving.

The policies are presented in Table 4 in a generally decreasing order of active government involvement from left to right, and grouped as mandatory, voluntary, or incentive-based. The strongest tools of local government were the elimination of outdated production facilities and a capacity cap for a given industry in a region. Both effectively disabled a firm's facilities or prevented a firm's entry with an energy-intensive production process. Indeed, most of the firms invested in new plants and equipment not because of direct requirements, but because of the concern of ineligibility under stricter elimination standards in the future, or the risk of being suspended to save the production capacity for new, more efficient firms to enter.

The local governments also scheduled electricity quotas and rolling blackouts in advance (see e.g. Fisher-Vanden *et al.* 2012 for the general background), which caused suspension of part or all of a firm's production facilities during a short period. The advanced schedules gave firms incentives to invest in electricity saving technologies to make sure that their facilities could still operate with reduced electricity consumption, or as bargaining power with the governments to incur fewer blackouts in the future. The local governments also mandated the use of certain technologies, set energy efficiency performance standards for products and processes, and proposed specific environmental standards beyond the national requirements. These requirements were closely followed by firms and often led to the

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Table 4 Policies that affected firms' energy-saving decisions

	Mandatory						Voluntary			Incentive-based		
	Expected facility elimination	Local capacity cap	Electricity quota or blackout	Technology mandate	Energy efficiency standard	Local environmental request	Energy saving agreement	Government communication	National policy agenda	Financing support	Investment subsidy or reward	Resource conservation incentive
Cement 1	X	X	—	—	—	—	—	—	—	—	—	—
Cement 2	X	X	X	—	—	—	—	—	—	—	—	—
Cement 3	—	X	—	—	—	—	X	—	—	X	—	—
Cement 4	—	X	X	—	—	—	—	X	—	—	—	—
Cement 5	—	—	X	—	—	—	—	—	X	—	—	—
Cement 6	—	—	X	—	—	—	X	—	X	—	—	—
Cement 7	X	—	—	X	X	—	—	—	—	—	—	—
Steel 1	X	X	—	X	—	X	X	X	—	X	X	X
Steel 2	X	—	—	X	—	X	—	—	—	—	X	—
Steel 3	X	—	—	—	X	X	X	—	—	X	X	—
Steel 4	—	—	X	—	—	—	—	—	—	—	—	—
Paper 1	X	—	—	X	—	—	—	—	—	—	—	—
Paper 2	—	—	—	—	—	—	X	—	X	—	X	—
Paper 3	—	—	—	—	—	X	X	X	—	X	—	—
Paper 4	—	—	—	—	—	—	X	—	—	—	X	—
Paper 5	X	—	—	—	—	—	—	—	—	—	—	—
Chemical 1	—	—	—	—	—	—	X	X	—	—	—	—
Chemical 2	X	—	—	—	—	—	X	X	—	X	—	X
Chemical 3	—	—	—	—	—	X	—	—	—	—	—	—
Chemical 4	—	—	—	—	—	—	—	—	—	X	—	—
Chemical 5	—	—	—	—	—	—	—	—	—	X	—	—

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adoption of new technologies. Mandatory instruments more frequently affected cement and steel firms, because of central policy setting and local regulatory needs to address overcapacity in the two industries.

Voluntary and incentive-based instruments required greater firm participation, but local regulators also played essential roles by overseeing that the policies were successfully implemented. These regulators expanded a national list of firms to be considered in the voluntary energy-saving agreement program by incorporating locally important energy consumers, and encouraging other firms to develop energy saving plans and sign energy saving agreements.[7] Informally, the regulators often expressed the needs of the local governments in energy saving, interpreted national policies, recommended energy saving technologies and practices, and kept close track of firms' practices.[8] They also provided generous financial support and additional incentives beyond the national standard.[9] Even when financing policies gave energy service companies tax relief to encourage them to invest in energy-saving performance contracts, local authorities attempted to facilitate the process by providing opportunities for energy service companies and industrial firms to meet and build on previously successful collaborations.[10]

The critical role of the local governments in implementing these policy instruments can be clearly observed by comparing across municipalities: with less need for energy saving and a relatively lower level of economic development in municipality D, firms (Cement 6 and 7, Steel 4, Paper 5) were little affected by the voluntary instruments and not affected by incentives at all, while firms in other municipalities were encouraged to save energy under the two types of policy instruments.

4.2. Enforcement responsiveness to the regulated

The authoritarian rulemaking and operationalization of local energy efficiency was complemented by responsiveness to the regulated firms in the enforcement process. Responsiveness was manifested most clearly in the process of enforcing the strongest mandatory policies: (i) eliminating outdated facilities and maintaining local capacity caps, and (ii) setting electricity quotas and blackouts. There were two types of regulatory responses for enforcement, one with direct preferential enforcement of less stringency compared to other firms, and the other alleviation of previous punitive enforcement. Responsive enforcement was applied to all cement firms, all but one steel firm, and two chemical firms (Table 5).

Among the firms under preferential enforcement, three out of eight managed to survive or expand under strict regulations on plants and equipment through negotiation and demonstration of good performance. One cement firm (Cement 7) replaced its small-scale vertical cement kilns expected to be on the elimination list in the future with larger ones and expanded its overall production capacity under the local capacity cap at the expense of the closure of other similar firms, because it demonstrated to the local government its competitiveness with land and human resources. Another cement firm (Cement 1) with a small-scale vertical kiln survived along with two others of its kind while hundreds of similar firms in Jiangsu Province were closed. The surviving firm updated its kiln to have higher energy efficiency, and more importantly, to use local mine tailings and industrial waste as its inputs. A steel firm (Steel 3) with a furnace on the elimination list managed to replace its old furnace with a new one just above the minimum standard, because it had been providing community service by making available by-products – furnace gas and coke oven gas – to local residents as fuel at a price lower than the market rate. Without a similar contribution to the local community, most other firms with facilities on the elimination list were either required to close or to install facilities much more advanced than the elimination standard.

Additionally, four of the interviewed firms enjoyed more generous electricity quotas or avoided rolling electricity blackouts, which meant that their regular production was minimally affected. According to the interview respondents, they negotiated and acquired the benefits because of their superior performance in energy efficiency and energy saving (Cement 5 and 6, Steel 2), leading environmental performance (Cement 6), and contribution to local tax revenue (Cement 5). As an exception, one firm (Chemical 1) did not even need to negotiate for secure electricity supply, as the government fully acknowledged the firm's essential role in the supply chain of the local chemical industry. One remaining firm (Chemical 2) was advised and endorsed by the government to develop a central energy information system, because of its advanced technical basis in energy information collection and management, which was acknowledged by the regulators through routine examinations.

Compared to preferential enforcement, greater interaction between the regulators and the regulated was observed in cases where previous regulatory decisions had been changed. A typical case involved two cement firms (Cement 1 and 2) in the same district of municipality A. Although Cement 1 survived through an extensive

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Table 5 Examples of enforcement responsiveness

Firm	Location	Enforcement responsiveness	Reason
Preferential enforcement			
Cement 1	A	Closure avoided	Community service, energy efficiency
Cement 5	B	Alleviated electricity quota	Tax revenue, energy efficiency
Cement 6	D	Alleviated electricity quota	Energy efficiency, environmental performance
Cement 7	D	Expansion approved	Energy efficiency, competitiveness
Steel 2	B	Alleviated electricity quota	Energy efficiency
Steel 3	C	Alleviated facility turnover	Community service
Chemical 1	A	Avoided electricity quota/blackout	Essential in supply
Chemical 2	B	Technology recommendation	Technology basis
Changing enforcement			
Cement 1	A	Closure → partial suspension	Employment
Cement 2	A	Expansion suspended → approved	Community service, environmental performance
Cement 3	A	Entry suspended → approved	Tax revenue, energy efficiency, environmental performance
Cement 4	B	Expansion suspended → approved	Community service, energy efficiency
Steel 1	A	Relocation → relocation avoided	Energy efficiency, environmental performance, employment

Enforcement responsiveness is featured either by changing enforcement with an arrow connecting two enforcement decisions, or by preferentially less stringent enforcement without an arrow. A–C represents three municipalities in the south and D represents a municipality in the north of Jiangsu Province. Environmental performance (such as treatment of municipal waste and fuel supply for residential use).

round of facility elimination, it was under the threat of closure again a few years later. The closure threat was related to the fact that Cement 2 applied to the government for construction of a new, advanced cement kiln with a larger capacity for cement clinker production (an intermediate blended into final cement products). This approval required closure of the same capacity of production, in aggregate, of other kilns in the same district.

While Cement 1 still used a large portion of mining and industrial waste, this contribution to the local environment was no longer essential, as many other firms could also reuse waste. In comparison, Cement 2 was able to treat and reuse municipal solid waste and water treatment sludge, which few other firms could manage. Cement 2 also proposed to treat hazardous waste in its cement kiln that other firms could not treat, and to be the first cement firm to install a NO_X reduction device, which would help the local government demonstrate and promote installation of similar devices to control NO_X. Because of the unique contribution that Cement 2 could make to local environmental management, its expansion plan was approved with the closure of cement kilns in the same district, including that of Cement 1. Cement 1, however, also made a strong case in negotiation that it had a large number of employees that could not be easily dismissed without damaging their well being. Therefore, the local government only banned the cement kiln at Cement 1 for cement clinker production but not production of final cement products, and helped to coordinate so that Cement 1 could get a stable supply of cement clinker from Cement 2.

Similarly, Cement 3 and 4 had their new cement kilns approved for construction after the government had initially suspended their expansion plans, through successful negotiation and commitment to energy saving, environmental protection, treatment of municipal solid waste, and contribution to local tax revenues. A steel firm (Steel 1) finally avoided the municipal government request to relocate after long-term discussion explaining the difficulty of accommodating its large number of employees and their families, and by explaining and committing to its future plans for using more advanced production facilities, investment in energy saving, and environmental protection.

4.3. Scale and scope of responsiveness under authoritarian rule

In addition to the presence of and conditions for enforcement responsiveness, we also investigated its scale and scope, particularly under authoritarian rule. The design of energy efficiency regulation includes a set of instruments with different degrees of reward or punishment. This makes it possible for regulators to use different

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instruments in combination, establishing a tit-for-tat strategy to improve compliance, following the idea of responsive regulation (Ayres & Braithwaite 1992). We find less evidence, however, of this strategy of large-scale responsiveness in regulation combining different policy instruments, and more evidence of small-scale adaptation in enforcing individual policy decisions formulated through the authoritarian process.

Unlike the ideal of responsive regulation (Ayres & Braithwaite 1992; Parker 2013), where regulators start from the base of the enforcement pyramid and presume the regulated to be cooperative, local regulators in Jiangsu Province enforced energy efficiency policies with a formal and coercive approach from the very beginning. This authoritarian rulemaking and operationalization helped to introduce energy saving as a new issue under regulation, reduce uncertainty of the progress of compliance, and achieve local policy targets on time. With better knowledge about firms' behaviors, attitudes, and plans, the regulators became more flexible and cooperative by altering enforcement levels and decisions, especially for mandatory policies that affected firms' survival or regular production. Such responsiveness, however, was still well managed by the regulatory body with strong power and coordination.

In contrast to a small scale of responsiveness to single enforcement incidents, the scope to which regulators were responsive was found to be broad, cutting across actions and attitudes on different regulatory issues. Energy saving was the regulators' main concern. Although some interviewees did not mention energy efficiency as a reason that their firms enjoyed preferential enforcement or had enforcement revoked, it was a necessary condition. All firms under responsive enforcement presented superior energy efficiency, high levels of energy saving, or both. Future plans and commitment for energy saving were also important considerations, especially for large firms that contributed to significant shares of local energy consumption.

In all but one case, however, performance in energy efficiency was not the single reason to persuade regulators to alleviate enforcement. The regulators and local authorities usually took other contributions that a firm could bring to the local community into consideration. These included demonstration of superior environmental performance and efforts; provision of services that treat local industrial, municipal, and hazardous waste; high employment levels; and contribution to local tax revenue. While these issues were usually managed by separate government agencies, the administrative coalition and broad participation of regulatory agencies on energy saving, that is, an authoritarian regulatory capacity, helped to coordinate between these different regulatory issues. Therefore, energy efficiency policies were used as leverage to promote compliance with other regulatory issues.

The scope of responsiveness varied across areas and changed through time, which reflects the demand of local authorities and suggests their responsive decisions as rational choices. While the regulators were practicing enforcement responsiveness, the scope of responsiveness and its changes were increasingly revealed to the regulated firms. Regulators in areas with a less developed economy, for example, were more responsive to firms that would make great contributions to local tax revenue (Cement 3 and 5), while regulators in or around urban areas were more responsive to firms that could provide services in waste management (Cement 1 and 2) and utilities (Steel 3). From 2006 to 2010, industrial waste reuse in municipality A gradually evolved from a contribution that cement firms made to the local community to a normal practice, because more firms realized the potential profit and policy preference regarding waste reuse (Zhu & Chertow 2016) and became involved in the business. During the same time, reuse of municipal solid waste and sludge from water treatment plants became an essential contribution cement firms could make. This service is also expected to become normal practice: one firm (Cement 2) has committed to treat hazardous waste as the new leverage for local authorities' lessened regulation on electricity quotas and rolling blackouts, as well as other measures that may affect production.

5. Discussion

The authoritarian but responsive approach to local energy efficiency strengthened regulatory capacity and policy implementation. Of course, the dedication to energy efficiency and environmental protection did lead to harsh results, such as closure of hundreds of firms. But with explicit regulatory targets and interactive processes, firms improved their understanding of local regulation and complied and contributed more actively to local governance. All four municipalities improved energy efficiency beyond their own targets without compromising economic growth.

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Our observations both resemble and differ from other research about transitions of the policy process in China. Authoritarian environmentalism has been increasingly studied in the literature characterized by quick response, government dominance, and limited public participation in the policy process (Beeson 2010; Gilley 2012). But unlike the concern over local collusion and deviation from the central government intention under authoritarianism (Zhou 2010; Gilley 2012), energy efficiency regulation has also been operationalized consistently at the local level in an authoritarian way, with clearly targeted delegation, powerful and coordinative leading groups, strategic plans, and multiple policy instruments and operational measures. The improved implementation has been similarly explained elsewhere as the result of local informal coalitions and strategic bundling of national priorities with local interests (Kostka & Hobbs 2012; Harrison & Kostka 2014), and campaign-style enforcement that strengthens the importance of environmental legitimacy and enhances economic incentives for environmental compliance (Liu *et al.* 2015).

While the target-based mandatory instrument has been particularly criticized in terms of its rigidity, inappropriateness to local contexts, and cycles of action and inaction (Kostka 2015), we find that the local authorities in all the studied municipalities in Jiangsu Province have been flexible and responsive to industrial performance when enforcing mandatory policies, leading to better compliance and improvement in energy efficiency. Responsive enforcement complements authoritarian implementation and improves the overall regulatory effect.

The responsiveness of energy efficiency regulation, however, differs from the literature of responsive authoritarianism and that of responsive regulation in terms of its scale, scope, and stage in the policy process. The current discussion of "responsive authoritarianism" in China features a transition to managed participation between numerous policy actors (Cai 2004; Mertha 2009; van Rooij *et al.* 2014), political deliberation (He & Warren 2011), and growth in grey areas intentionally left out of governance (Weller 2012; Stockmann 2013). The responsiveness of county officials to citizen requests increases with both the threat of collective action from the bottom and complaints that trigger oversight from upper-level governments (Chen *et al.* 2016). While our findings show similar responsiveness to the increasingly diverse demands of Chinese society, such responses are more focused, occurring only in the enforcement stage of specific regulation and being limited to the targeted group of regulated firms. For the same reason, enforcement responsiveness clearly differs from the decentering of regulation, where state-control becomes fragmented.

The responsiveness is similar to "responsive regulation" (Ayres & Braithwaite 1992; Parker 2013) in terms of regulators' adjustment of enforcement stringency according to the behavior and attitudes of the regulated (Mascini 2013), based on direct face-to-face interactions on manageable projects (Ford 2013). The enforcement responsiveness we observed, however, differs from this "responsive regulation" ideal in its small scale of responsiveness, often on single policies, and on its starting point of enforcement being authoritarian and formalistic, instead of a presumption of collaboration and self-regulation. It also differs from "responsive regulation" by having a broader scope of responsiveness to regulatory issues beyond energy efficiency. Both the narrow scale and the broad scope of responsiveness is determined by the other part of the approach, authoritarian operationalization, which makes the initial enforcement more formalistic and coercive, but also builds strong regulatory capacity to coordinate across multiple regulatory issues.

Because of the authoritarian logic that guides enforcement responsiveness, this approach also differs from enforcement styles discussed in the literature. The choice of enforcement styles is often at the discretion of street-level bureaucrats or agencies for single regulatory issues (May & Burby 1998; May & Winter 2000). In contrast, the authoritarian but responsive approach of local governments in China emerged under China's decentralized authoritarian regime. The issue of energy efficiency was taken up by the central government over a short period, featuring top-down delegation of a very aggressive policy target. Because energy consumption statistics were relatively transparent and built into the central policy design, energy saving progress was closely monitored, evaluated, and linked to local cadre evaluation and promotion.

Given the strong political interest in reaching the policy targets assigned by the national government, on the one hand, local authorities had to follow the central policy closely and rely on an authoritarian approach to guarantee immediate improvement. On the other hand, regionally-based economic and administrative governance enabled local authorities to explore alternative policy measures and required them to take care of other issues regarding local development and welfare. In this way, local authorities were able to rationally lessen and change enforcement decisions to accommodate the needs of the regulated firms, in order to encourage better

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performance not only in energy saving, but also in other issues, such as environmental protection, employment, and fiscal revenue.

This approach has its limitations and strengths. According to Braithwaite (2006), responsive regulation is a democratic ideal, where new problems, forces, and demands are taken into consideration; accountability is held by several separated powers; and deliberation and responses are practiced in social circles of everyday life. The responsiveness we observed in energy efficiency regulation is far from such a democratic ideal, with responses to the demands of the regulated firms predominantly controlled by the regulators with limited public participation in the policy process. The more sophisticated regulation under the decentralized authoritarian system resembles what has been called "consultative authoritarianism" for the state-society relationship, with limited potential deliberation toward democratization (Teets 2013).

Too much responsiveness, however, may lead to favoritism, if local governments can easily reach their policy targets and be more selective in enforcement based on personal and political connections. While favoritism from individual regulators in enforcement has been suppressed so far by a broad engagement of government agencies in authoritarian rulemaking and operationalization, further elimination of the potential risk requires the mechanism of responsiveness to be more routinized and under interrogation.

Nevertheless, in the case of energy efficiency regulation where local governments have to achieve policy targets on time, the regulatory approach expands the options beyond the previous cases of lack of enforcement or purely authoritarian environmentalism in China. The local authoritarian rulemaking and operationalization features strengthened regulatory capacity, quick response, multiple instruments applied to different firms, and therefore the capability to develop greater policy effectiveness. The responsive process helps firms understand and adapt to local regulation, leading to better compliance. Indeed, it is important to note that the local authorities oversee multiple dimensions of firms' performance and respond broadly, even beyond energy efficiency, to issues of environmental protection and local welfare. They respond in a calculated manner to coordinate across multiple and sometimes overlapping regulatory goals. This helps to optimize local regulation and to overcome the common problem of incomplete regulation, where too few issues make it onto the government agenda to be formulated into policies. In particular, it may mitigate negative impacts from a heterogeneous disaggregation of policy targets in China (Zhang *et al.* 2013).

Responsiveness in China is triggered, in part, by the needs of a rapidly industrializing economy facing collective problems, such as pollution abatement and energy accessibility. We have seen a pattern, over time, of the passage of a series of laws over a multi-year period as problems are better understood and interpreted. The drive for a law to urge cleaner industrial production (law implemented in 2003), for example, came to be seen more broadly as a need for a "circular economy" involving production and consumption (law implemented in 2009) and, more recently, for an inclusive systems view of an "ecological civilization." When a new national law is promulgated and enacted it does not, however, eliminate those that came into effect earlier, exacerbating the problem of overlap described above. Resolution depends upon the extent to which local governments have previously taken up enforcement within specific policy contexts, such as cleaner production, and where they are on the agenda.

Given the advancement of the authoritarian but responsive approach in local regulation, the extent to which the approach has been applied, and the context for facilitating the approach are of practical importance. The geographical context of our research, Jiangsu Province, represents a region with a highly developed economy and strong government capability. In another well-developed city, Guangzhou, research shows a trend toward a more formalistic and coercive approach by the officials in enforcing environmental policies, rather than a collaborative or responsive approach (Lo *et al.* 2009). Given the extra resources and steps involved in being authoritarian but responsive, regions with lower social and economic development levels and less regulatory capacity would also be less likely to apply the approach.

Participating in the authoritarian but responsive approach requires both strengthened local regulatory capacities and improved central policy designs. Energy efficiency policy successfully combined transparent policy targets, energy statistics, and clear accountability with choices of multiple policy measures and instruments, so that local authorities had strong political interests in saving energy and technical feasibility for monitoring and enforcement, but could also maintain flexibility in implementation. If future policy objectives could be well formulated and linked to local evaluation, enforcement responsiveness might become more likely to be adopted for local optimization over regulatory issues.

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6. Conclusion

Understanding the capacity, strategy and effectiveness of local policy implementation and enforcement has been an important theme in the literature. Here we find the authoritarian but responsive approach to local regulation, as illustrated by energy efficiency in Jiangsu Province, to be a useful model of description and explanation. Local regulators formulated and operationalized the policy in an authoritarian way, by building regulatory capacity with strong power and coordination, designing strategic plans, and actively applying multiple policy instruments beyond the national framework. Later on, they responded to local firms with more tailored enforcement procedures by applying preferential decisions or alleviating punitive ones. Enforcement responsiveness prevents authoritarian rulemaking from being too inflexible, thereby preempting unintended policy consequences.

In the case where numerous laws in specific policy arenas are being passed down to the local level, enforcement responsiveness provides the local official an opportunity to try to harmonize and adjudicate across many federal directives to lessen confusion among constituencies, especially companies with increasing financial and operational risks. Authoritarian rulemaking and operationalization confines the scale of enforcement responsiveness but broadens its scope to issues other than energy efficiency through an empowered regulatory body that can coordinate as needed. This approach not only encourages firms' compliance with energy efficiency, but also their contribution to environmental protection and local governance.

Recognition of the authoritarian but responsive regulatory model offers a consistent explanatory framework for seemingly fluctuating or discretionary enforcement incidents. The rationale is embedded in the regionally decentralized authoritarian system in China, where local officials are politically responsible to higher authorities for social and economic development, but can apply flexible approaches. On the one hand, local officials apply this approach to strengthen and coordinate across different regulatory issues, which improves local governance and ultimately benefits their own careers. On the other hand, responsiveness is restricted in the regulatory process, follows an authoritarian logic, and does not necessarily imply a deliberation and democratization process, or even a decentering of regulation among new political actors. This approach differs from the conceptualization of responsive regulation or responsive authoritarianism in its scale and scope of responsiveness. It also has advantages over the inflexible authoritarian environmentalism and highly variable shifts of enforcement styles.

Much work remains to be done to explore the extent to which this approach has been used, the detailed contextual factors to support the approach, and the conditions for it to be a good or bad regulatory approach. It is also important to understand the long-term normative implications of the dynamic interactions between the regulators and the regulated.

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Notes

1 Government interviews, province #1, municipality A #1, and municipality B #1.
2 Government interview, province #2.
3 Government interviews, municipality A #1, municipality B #1 and #2, municipalities C and D.
4 Government interview, municipality A #2.
5 Government interviews, municipality B #1 and #2.
6 Government interview, municipality A #1.
7 Government interview, municipality A #1 and municipality B #1.
8 Government interview, municipality B #2.

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9 Government interview, municipality C.

10 Government interview, municipality B #1.

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Appendix

Recent industrial energy efficiency in China

This appendix reviews the major guidelines and detailed policy programs related to industrial energy efficiency in China between 2004 and 2014. The major guidelines and laws related to energy saving include:

1. *Medium and Long Term Energy Saving Plan* by the National Development and Reform Commission in 2004.
2. *Notice for Building a Conservation-oriented Society* by the State Council in 2005.
3. *Outline of the 11th Five-Year Plan for National Economic and Social Development* by the National People's Congress in 2006.
4. *Decisions on Strengthening Energy Saving* by the State Council in 2006.
5. *Comprehensive Work Plan for Energy Saving and Emission Reduction* by the State Council in 2007.
6. *Energy Conservation Law Amendments* by the National People's Congress in 2007.
7. *Outline of the 12th Five-Year Plan for National Economic and Social Development* by the National People's Congress in 2011.
8. *Comprehensive Work Plan for Energy Saving and Emission Reduction in the 12th Five Year* by the State Council in 2011.
9. *Work Plan for Controlling Greenhouse Gases Emission in the 12th Five Year* by the State Council in 2011.
10. *Energy Saving and Emission Reduction Plan in the 12th Five Year* by the State Council in 2012.
11. *Industrial Energy Saving Plan in the 12th Five Year* by the Ministry of Industry and Information Technology in 2012.
12. *Industrial Response to Climate Change Action Plan 2012–2020* by the Ministry of Industry and Information Technology, the National Development and Reform Commission, the Ministry of Science and Technology, and the Ministry of Finance in 2012.

The industrial energy saving measures proposed in these guidelines are listed in Table A1. These energy saving measures are implemented through detailed policy reforms and programs, which are selectively listed in Table A2.

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Table A1 Industrial energy saving measures in major law and guidelines from 2004

Year	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Target and performance setting												
National energy intensity target	X		X	X	X		X	X		X		
National carbon intensity target							X		X			
Industrial energy intensity target										X	X	
Industrial carbon intensity target												X
Target disaggregation				X	X			X		X	X	X
Local energy saving plan					X	X		X				
Low-carbon development plan									X			X
Local official examination			X		X	X	X	X	X	X		
Process/product energy standard	X	X		X	X	X		X		X	X	
Local industry process energy standard					X	X					X	
Industrial structure adjustment												
Guidance for structural adjustment	X			X	X			X	X	X	X	
Outdated capacity elimination	X	X	X	X	X	X		X	X	X	X	X
Project energy saving assessment			X	X	X	X	X	X	X	X	X	
Land and loan constraint					X			X		X		
Capacity constraint					X		X	X	X	X	X	X
Targeted administration												
Energy intensive industries	X	X	X	X	X	X		X	X	X	X	X
Key energy consuming firm	X	X	X	X	X	X	X	X	X	X	X	X
Firm energy saving plan and report					X	X		X			X	X
Firm energy saving audit				X	X	X		X		X	X	
Monitoring and statistics				X	X	X		X		X	X	
Technology and information												
Key energy saving projects	X	X	X	X	X		X	X	X	X	X	X
Energy saving equipment catalog	X		X	X		X	X	X	X	X	X	X
Energy saving technology promotion	X	X				X		X	X	X	X	X
R&D support	X	X		X	X	X	X	X		X	X	X
Comprehensive resource utilization									X		X	X
Energy information system										X	X	X
Education	X	X		X	X	X		X	X	X	X	X
Supporting policy reform												
Energy price and tax reform	X	X	X	X	X	X	X	X		X	X	
Electricity pricing	X	X	X	X	X	X	X	X		X	X	
Energy saving financing support	X	X	X	X	X	X		X		X	X	X
Energy saving tax breaks		X	X	X	X	X		X		X	X	
Export tax and quota		X			X	X	X	X		X	X	X
Energy saving reward				X	X	X		X		X	X	
Energy saving subsidy					X	X		X		X	X	
Demand side management	X	X	X	X	X	X	X	X		X	X	
Energy saving performance contract	X	X	X	X	X	X	X	X		X	X	
Promoting energy service industry	X			X	X	X	X	X	X	X	X	X
Voluntary agreement	X					X					X	X
Greenhouse gases emission accounting									X			X
Carbon emission trading								X	X			X
Low-carbon trial site									X			X

R&D, research and development.

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Table A2 Selected national policies and programs for industrial energy saving from 2005

Year	Agency	Policies and programs
		Target setting and tracking
2006	SC	Regional Energy Intensity Reduction Targets in the 11th Five Year Plan
2012	NDRC	Notice on Regular Disclosure of Energy-Saving Target Achievement Barometer
		Firm energy saving administration
2006	Multiple	Top-1000 Enterprise Energy Saving Program Implementation Plan
2006	NDRC	Assessment Guideline for Enterprise Energy Audit Report
2006	NDRC	Assessment Guideline for Enterprise Energy Saving Plan
2007	NDRC	Energy Saving Assessment and Review Guideline for Fixed-Asset Investment
2008	NDRC	Implementation Plan for Key Energy-Consuming Enterprises Energy Use Reporting Scheme
2008	Multiple	Energy Saving Audit Guideline for Energy Saving Projects
2010	SASAC	Interim Measures on Supervising and Administrating Energy Saving in State-owned Enterprises
2011	Multiple	Top-10000 Enterprise Energy Saving Low Carbon Program Implementation Plan
		Technology promotion
2006	Multiple	Implementation Plan for Ten Key Energy-Saving Projects in the 11th Five Year Plan
2008	NDRC	National Energy-Saving Technology Promotion Catalog
2009	MIIT	Recommended Catalog for Energy-Saving Electric Equipment (Products)
2012	SC	Energy-Saving Environmental Protection Industry Development Plan in the 12th Five Year
2014	NDRC	Interim Measures on Promoting Energy-Saving and Low-Carbon Technologies
		Selected industrial policies
2005	SC	Interim Provisions on Promoting Industrial Structure Adjustment
2005	NDRC	Guiding Catalog for Industrial Structure Adjustment
2005	NDRC	Iron and Steel Industry Development Policy
2005	NDRC	Salt Industry Structural Adjustment Guidance
2006	NDRC	Cement Industry Development Policy
2006	Multiple	Opinions on Promoting Structural Adjustment and Upgrading of Textile Industry
2007	NDRC	Paper Industry Development Policy
2008	NDRC	Dying and Printing Industry Entering Policy
2009	Multiple	Opinions on Preventing Overcapacity and Redundant Construction in Some Industries and Guiding Healthy Industrial Development
2010	SC	Notice on Further Elimination of Outdated Production Capacity
2010	MIIT	Guiding Catalog for Eliminating Outdated Production Technologies, Equipment and Products in Some Industries
2013	MIIT	Opinions on Guiding Energy Saving and Emission Reduction in the Nonferrous Metals Industry
		Selected incentive-based policies
2005	NDRC	Interim Measures on Central Government Budget for Project Investment and Interest Subsidy
2006	Multiple	Recognition for Nationally Promoted Resource Comprehensive Utilization
2007	Multiple	Interim Measures on Financial Reward for Energy-Saving Technological Transformation
2007	Multiple	Interim Measures on Central Government Financial Rewards for Eliminating Outdated Production Capacity
2008	Multiple	Energy-Saving and Water-Saving Equipment Catalog for Corporate Income Tax Benefit
2010	Multiple	Opinions on Promoting Energy-Saving Performance Contract and Development of Energy Service Industries
2010	Multiple	Interim Measure on Financial Rewards for Energy-Saving Performance Contract
2010	Multiple	Notice on Value-added Tax, Sales Tax, and Corporate Income Tax Policy for Promoting Energy Service Industry

MIIT, Ministry of Industry and Information Technology; NDRC, National Development and Reform Commission; SASAC, State-owned Assets Supervision and Administration Commission; SC, State Council. Collected from the agencies' websites.

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